

07 — Graded Year 2 entry sheet (Plan 2.2)

Status: graded run, Year 2 of 4. Opening position is the Year-1 REP balance sheet (p6): cash 46,902,208, receivables 27,284,077, payables 15,380,390, loan 26,000,000 (6,500,000 due 2 January), H finished goods 17,628,699 units at 2.3252 (40,990,251), S raw material 298,350 units (507,046), 10 S + 22 H lines (54,000,000 gross), rank 1 of 5 on 67,823,192 of net income. Written 2026-09-06. Every figure traces to a § of Scenario MMT39 EN v231.pdf , a Year-1 result file under day 2/year 1 results/ , the test-round Year-2 files, or the scratch script plan_y2.py (imports model/mmt39.py read-only and applies the twelve measured corrections of day 2/year 1 results/analysis/03-model-vs-actual.md by hand; model/mmt39.py is untouched except loan_maturity_years 3 -> 4).

Markers: [!] trap · [+] edge · [-] judgement call. No emoji inside tables.

One correction to the brief before anything else. [!] The strategy this sheet was asked to quantify added four S lines (10 -> 14). That is not a legal entry: \$22 caps the factory at **32 lines in any S/H combination**, and 10 + 22 = 32 is already there (SCENARIO0["max_total_lines"] ; Doc 06 build path: "None left: the \$22 cap is reached"). Lines can be switched off, never removed or converted (\$22), so idle H lines free no slots. The only S-capacity lever left is **SMED** (\$27, first available in year 2), measured in the test round at **+14.1% S / +14.4% H output per line-shift** (222,500 / 557,500 against 195,000 / 487,500; Doc 05 Part 4). Everything below is built on SMED instead of new lines, and two consequences follow: S stays supply-capped below last year's measured demand, so the B promotion the brief proposed is pointless (\$48 rejects promotions the supply cannot serve) and the S price becomes the lever; and the 6,000,000 of SMED capex replaces 8,400,000 of line capex in every cash figure.

Decisions taken by the team and kept as stated: sell the 17.6M H stock before making more; H at 7.50 in both markets with a price-reduction promotion in supermarkets; hold S media and campaigns; buy the survey set that explains the H loss in A; drop the training department; no new loan, prepay the old one if the trough allows; no E in Year 2.

The sheet in ten lines

#	Item	Value	Source
1	Lines	No new lines (cap reached). 10 S active, 2 of 22 H active. Capex: 2 SMED modules, 6,000,000 , 2 teams per shift	\$22 cap; \$27; test Y2 DEP p6 ran exactly 2 modules / 2 teams
2	Shifts / maintenance	3 shifts , preventive ON, Poka Yoke OFF (720,000 x 32 lines = 23,040,000, \$26), training department OFF	\$23, \$25, \$26, \$32; Form 3
3	Planned output	S 6,675,000 · H 3,345,000 (222,500 / 557,500 per line-shift with SMED); H available for sale 17,628,699 + 3,345,000 = 20,973,699	Doc 05 Part 4 measured per line-shift; REP p3 opening stock
4	Raw material to buy	S 6,578,235 · H 3,446,019 units, both on cash terms (1.02 x output x 1.01 buffer, less the 298,350 S units in stock)	REP p4 yield 1.0200; Form 7 note; Form 8 terms table

5	Prices (MSRP), all channels	S 15.00 in A and B (was 14.50) · H 7.50 in A and B (was 6.50 / 7.50) · margins S 1.90 / H 1.00 · shelf S 18% / H 22%	Section 2.2 (S supply-capped 1,002,440 below Y1 demand); Doc 02 brief #1 for H
6	Promotions	H: price reduction, 3 promotions, 0.50, supermarket channel only, A and B. S: none (supply-capped). Cost 661,702 on ~1,323,403 promo units	\$18; \$48; survey 22 (Co2 and Co4 both promoted and both sold out); section 2.3
7	Sales network / fleet	6 / 6 / 0 managers; reps A 4/4, B 5/3 at 27,000 + 0.60%. Platforms 12 / 12 / 12. Trunk 2 A, 2 B @ 24,000 kg; last mile 24 A, 18 B @ 10,000 kg	\$6, \$10-\$13; section 2.7
8	Media / surveys	Campaign 18 S / 19 H; S 8 TV + 4 PR, H 5 TV + 3 PR per market (678,000); POP 10,000 S / 15,000 H per channel (150,000). Surveys: 22 of 28, 680,600	--media: TV cheapest at 300 per reach point; Form 4
9	Finance	No loan. Prepay 19,500,000 (penalty 195,000). No credit line (saves 200,000). Fixed-term deposit 10,000,000 at 2.5%. Customer terms 60 days	\$42-\$44; INF p2; January trough 9,257,578 after all three (section 4.2)
10	Expected Year-2 net income	87,502,697 at last year's measured demand (S 7,677,440 capped to 6,675,000; H 15,047,394). Year-end cash 108,292,278 + 10,000,000 deposit; H stock left 5,926,305	plan_y2.py base case; Year 1 actual 67,823,192 for scale

[!] Line 10 credits nothing to the H promotion, nothing to the A price rise (up or down), and nothing to the S price rise: volumes are last year's measured demand for our brand (sold + "units demanded of your brand", REP p2-p3). The S line is capped by output, so every S column of the sensitivity table above 6,675,000 reads the same. The H stock left (5,926,305) is 4.9 months of Year-1 H sales; it does not spoil (\$3) and sells in Year 3 before any new H production is needed.

1. Year-2 forecast, base case

Hand-corrected model, every correction from 03-model-vs-actual.md applied. Y1 actuals alongside.

Line	Year 1 actual (REP)	Year 2 base	How it is computed
Units produced S / H	5,850,000 / 32,175,000	6,675,000 / 3,345,000	10 x 3 x 222,500; 2 x 3 x 557,500 (test-Y2 SMED per line-shift)
Units sold S / H	5,850,000 / 14,546,301	6,675,000 / 15,047,394	S = output (demand 7,677,440); H = Y1 sold + unserved

Unserved demand S / H	1,827,440 / 501,093	1,002,440 / 0	S still short by a million units at Y1 demand
Sales at MSRP S / H	84,825,000 / 104,540,764	100,125,000 / 112,855,455	6,675,000 x 15.00; 15,047,394 x 7.50
Sales at MSRP total	189,365,764	212,980,455	
Retailer margins S / H	11,115,000 / 14,546,301	12,682,500 / 15,047,394	1.90 / 1.00 per unit
Net revenue	163,704,462	185,250,561	
Direct production cost S / H	32,528,340 / 33,823,059	34,370,243 / 34,760,649	S at 5.1491; H at the \$33 weighted average 2.3101 (17,628,699 @ 2.3252 + 3,345,000 @ 2.2305)
Cost of sales promotions	0	661,702	0.50 x 1,323,403 promo units (section 2.3)
Logistics (trunk + warehouse + last mile)	1,632,862	1,501,528	105,600 + 639,928 + 756,000 (section 2.7; vehicles at the measured 50% of \$10)
Gross profit	95,720,201	113,956,440	
Depreciation	6,480,000	7,200,000	12% x (54,000,000 lines + 6,000,000 SMED); land excluded (\$35, REP p5)
General overhead	600,000	600,000	\$29
R&D	8,521,459	9,584,120	4.5% x MSRP sales (\$36, reproduced to the euro in Y1)
Sales network A + B (fixed)	3,192,000	3,192,000	12 x 50,000 + 96 x 27,000
Sales force incentives	702,750	740,022	0.6% x (MSRP - margin) x normal units in channels S and T (03 fix 12)
Market research	166,800	680,600	Form 4
Advertising incl. POP	828,000	828,000	678,000 + 150,000, unchanged
Office and management staff	2,818,000	2,818,000	\$29 at Y1 salary levels
Preventive maintenance	500,000	500,000	\$25
Selection and training	722,000	0	Dropped (Form 3)

Turnover (office + sales)	306,000	366,000	4 x 24,000 with TS off + 9 x 30,000 (03 fix 9)
Total general expenses	24,837,009	26,508,743	
Operating income	70,883,192	87,447,697	
Loan interest + upfront fee / prepayment penalty / credit line	2,860,000 / 0 / 200,000	0 / 195,000 / 0	19,500,000 prepaid 2 Jan at 1% (INF p2); no line. Y1: REP p5 loans 2,600,000 + fees 260,000; line 100,000 + fee 100,000
Deposit income	0	250,000	10,000,000 x 2.5% (\$44, INF p2)
Net income	67,823,192	87,502,697	
Cash 31 Dec / fixed-term deposit	46,902,208 / 0	108,292,278 / 10,000,000	Reconciliation in 4.3
Receivables / payables	27,284,077 / 15,380,390	30,875,094 / 0	2/12 of net revenue (03 fix 5); cash terms on raw material
H finished goods left, units / value	17,628,699 / 40,990,251	5,926,305 / 13,690,353	20,973,699 - 15,047,394, at 2.3101
Loan balance 31 Dec	26,000,000	0	

Unit-cost decomposition the forecast uses (3 shifts, preventive, SMED; cash terms on raw material):

Component	S no SMED (Y1 actual)	S with SMED	H no SMED (Y1 actual)	H with SMED	Rule
Raw material	1.7335	1.6830	1.1031	1.0710	list 1.65 / 1.05 x 1.02 yield x terms premium (3% at 120 days, 0 cash)
Labour	3.3231	2.9124	0.9323	0.8152	crew x salary / 195,000 or 222,500 (H 487,500 / 557,500)
Repairs	0.0621	0.0545	0.0249	0.0217	model repair charge on the measured output
Power	0.4417	0.4417	0.2650	0.2650	INF rate x 1.767 (03 fix 2)
SMED teams	0	0.0575	0	0.0575	2 teams x 3 shifts x 4 specialists x 24,000 = 576,000 over 10,020,000 units (\$27, \$30)

Total	5.5604 (REP p4)	5.1491	2.3252 (REP p4)	2.2305	
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[~] The SMED team line is the conservative reading: the test-Y2 conversion cost (\$ 3.4276, H 1.1046, Doc 05) implies only ~206,000 of team cost across 45.5M units, not 576,000. The difference is 370,000 of net income the forecast leaves on the table.

1.1 Sensitivity: H demand x S demand

Base plan held fixed; only demand moves. Net income / S stock at year end / H stock at year end.

H demand \ S demand	6,000,000	6,675,000 (= S output)	7,677,440 (Y1 demand) and above
12,000,000	NI 71,215,742 · S 675,000 · H 8,973,699	NI 76,077,759 · S 0 · H 8,973,699	NI 76,077,759 · S 0 · H 8,973,699
15,000,000	NI 82,462,996 · S 675,000 · H 5,973,699	NI 87,325,013 · S 0 · H 5,973,699	NI 87,325,013 · S 0 · H 5,973,699
18,000,000	NI 93,710,250 · S 675,000 · H 2,973,699	NI 98,572,267 · S 0 · H 2,973,699	NI 98,572,267 · S 0 · H 2,973,699
21,000,000	NI 104,858,899 · S 675,000 · H 0 (26,301 unserved)	NI 109,720,916 · S 0 · H 0 (26,301 unserved)	NI 109,720,916 · S 0 · H 0 (26,301 unserved)

[+] The S columns 8,200,000 and 9,000,000 asked for in the brief are identical to the 7,677,440 column (output caps sales at 6,675,000; unserved becomes 1,525,000 and 2,325,000). [!] The only scenario that breaks the plan is H demand below ~12M (a repeat of the A collapse in both markets): net income still clears 71M, cash still clears 86M, and the H stock carried is 9.0M units. Every scenario ends with zero overdraft (section 4).

2. Alternatives, quantified

Each table changes one thing against the base case (net income 87,502,697). Deltas are Year-2 net income unless stated; capex is cash on 2 January (\$37).

2.1 S capacity: nothing / 1 SMED module / 2 SMED modules

Option	Capex	S output	Sold at Y1 demand	Unserved	Unit cost S	Net income	Delta
No SMED (Y1 plant)	0	5,850,000	5,850,000	1,827,440	5.5100	80,010,167	-7,489,530
1 SMED module	3,000,000	6,675,000 (if one module)	6,675,000	1,002,440	5.1491	87,862,697	+3,489,530

		serves 12 lines)					
2 SMED modules (base)	6,000,000	6,675,000	6,675,000	1,002,440	5.1491	87,502,697	0
3rd module	3,000,000	+26,880 S at most	-	-	-	-	~ -36

[+] SMED pays for itself inside the year: +825,000 S units at 7.2151 of contribution (5.95M) plus 0.36 per unit of lower labour on every unit (2.4M), against 720,000 of depreciation and 576,000 of team payroll. Cash at 31 December is within 520,000 of the no-SMED case because the extra units are sold and collected. [-] One module is the same output only if a single duplicate module can serve the 12 active lines; the test measured two modules on 32 lines, so there is no evidence for one. The 360,000 is not worth the risk on an irreversible purchase: open question 2. The third module was rejected in Doc 05 (residual 26,880 S units, 15-year payback). Poka Yoke would cost 23,040,000 (720,000 x 32 installed lines, \$26) for +4.55%: never.

Rival capacity and our A volume. [-] Co2 (6 S lines implied) and Co4 (8) both sold out in A at lower prices (survey 3, 28; Doc 02 brief \$2.7). Their A sunscreen volume was 5,176,180 units. Market A potential is 12,100,000 (\$3 x \$2) against 10,935,676 sold, so 1,164,324 of potential was unserved alongside our own 1,827,440 of unserved demand. If rivals add 30% (+1,552,854 units) that lands entirely on the unsold potential plus the demand we could not serve anyway; our 6,675,000 is untouched. At +50% (+2,588,090) about 421,000 units of ours are at risk if every displaced buyer leaves the highest-priced brand: -3.0M of net income at 7.2151 per unit, still above the no-SMED case. B (9.0M potential, 25% penetrated) absorbs the rest. The number that would change this is A demand for our brand at 15.00, which nobody has measured: survey 19 is on Form 4 for exactly that.

2.2 S price at capped volume

S price (A and B)	Contribution per unit	Net income at 6,675,000 sold	Delta vs 14.50	Volume at which it breaks even vs 14.50	vs Y1 demand 7,677,440	Marker
14.50 (Y1, brief)	6.7399	84,326,467	0	-	stocks out by 1,002,440	[!]
15.00 (base)	7.2151	87,502,697	+3,176,230	6,235,393	demand must fall 18.8%	[-]
15.50	7.6903	90,678,927	+6,352,459	5,850,112	demand must fall 23.8%	[-]
16.00	8.1655	93,855,156	+9,528,689	5,509,673	demand must fall 28.2%	[-]

13.00 in B only	4.9535 in B	87,353,136	-149,561	+26,068 B units needed	B sold 90,504 in Y1	[~]
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Contribution = price - 1.90 margin - 5.1491 - 4.5% R&D - 0.6% commission on the S+T share. [~]

The brief said hold 14.50 because the market took everything at the highest price in the industry. That reasoning cuts the other way once supply is fixed a million units below demand: at 14.50 we stock out again and take \$34 brand damage for a second year; at 15.00 we earn 3.18M more unless demand for our brand falls by a fifth, and the step is 3.4% on a brand that outsold Co2 (11.50-12.50) four to one at a 2.00-3.00 premium. 15.00 is the base; 14.50 is one number on Form 5 if the team disagrees. [~]

13.00 in B: the loss on the Y1 B volume is 149,561 and it breaks even at +26,068 B units, but with zero spare supply the extra B units would be A units sold for 2.00 less. Not this year.

2.3 H lines active, price and promotion

H lines active	H produced	RM to buy	Cash into new stock (at 2.2305)	At demand 15.0M: NI / stock end	At 18.0M: NI / stock end	At 21.0M: NI / unserved
0	0	0	0	86,906,214 / 2,628,699	96,721,733 / 0 (371,301 unserved)	96,721,733 / 3,371,301 unserved
2 (base)	3,345,000	3,446,019	7,461,000	87,325,013 / 5,973,699	98,572,267 / 2,973,699	109,720,916 / 26,301 unserved
4	6,690,000	6,892,038	14,826,000	87,644,617 / 9,318,699	98,936,584 / 6,318,699	110,228,552 / 0

[+] Net income barely moves with the line count at any demand up to 17.6M (the new units are slightly cheaper than the stock, 2.2305 vs 2.3252, which is the whole difference), so the choice is cash tied up against the cost of a stockout. Two lines put 7.5M into stock that we already know how to sell (3,345,000 units is 2.8 months of Y1 sales) and lift the ceiling from 17.6M to 21.0M; at 18M demand zero lines loses 1.4M of contribution and takes \$34 damage, at 21M it loses 12.8M. Four lines double the cash for a ceiling nobody has evidence for. **Idle lines cost depreciation and nothing else:** \$35 depreciates every installed line at the plant's shift rate (12%), so the 20 idle H lines carry 3,600,000 a year whether they run or not; crews are hired per active line-shift (\$30: production headcount falls from 3,144 to 1,404 + 24 SMED specialists); preventive maintenance is a flat 500,000 (\$25).

ASSUMPTION : \$22-\$25 name no other charge for an installed-but-inactive line and the Y1 REP has no line-level cost to test it against.

H price (A / B)	Net income at base volume	Delta	What it takes to break even	Marker
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7.50 / 7.50 (base)	87,502,697	0	-	[~]
7.00 / 7.50	85,265,139	-2,237,558	+705,733 A units (+15.0%) at 3.3394 of contribution	[~]
6.50 / 7.50 (Y1)	83,027,581	-4,475,116	Y1 evidence: 6.50 took 22.6% of A behind Co4 (6.50-7.20 with 3x2) and Co2 (9.50-10.50)	[!]
8.00 / 8.00	94,653,239	+7,150,542	Volume must hold within -11.1% (Co2 sold out at 9.50-11.00 with our campaign and positioning)	[~]

Contribution per H unit in A: 3.8139 at 7.50, 3.3394 at 7.00, 2.8649 at 6.50. [~] The cheapest H in A lost to the dearest; the surveys on Form 4 exist to find out why before the price is used as a lever. 7.50 in both markets is the price with a 55.8% share behind it (B).

H promotion (price reduction, \$18)	Promo units (est.)	Cost	Delta vs base	Marker
None	0	0	+610,089	[!]
1 promotion, supermarkets, 0.50	441,134	220,567	+406,726	[~]
3 promotions, supermarkets, 0.50 (base)	1,323,403	661,702	0	[+]
3 promotions, all three channels, 0.50	2,604,357	1,302,178	-605,332	[~]
3 promotions, supermarkets, 1.00	1,323,403	1,323,403	-661,702	[~]
3x2 instead (Co4's type)	1,323,403	3,308,508	-2,646,806	[!]

How many and what it costs: \$18 allows up to 3 promotions per channel per product per year, always of one type in a channel, each effective for 3 of 52 weeks; the synoptic overview defines the cost as the promotion cost times the units sold on promotion. Enter **Number 3, Type "Price reduction" (survey-22 code 2; the model's promotion_cost_per_unit(kind=1, ...) , Detail 0.50**. Estimated promo units = supermarket H volume x 9/52; supermarkets were our largest H channel in both markets (A 1,534,920, B 5,856,779, REP p3). Cost per promo unit 0.50 (--promotions 7.50); 3x2 would be 2.50. [+] The promotion is the one H lever with evidence behind it: both rivals that sold out promoted (survey 22), we did not, and \$48 is no obstacle with 20.97M units available. No lift is credited in the forecast, so the 661,702 is the whole downside. [~] All-channel and 1.00 versions are on the table for the team; the base keeps the spend where the volume is.

2.4 Supplier terms, loan, credit line, training

Supplier terms (S and H)	RM premium	RM cost	Net income	Delta vs base	January trough (with	Payables 31 Dec

					prepay + deposit)	
Cash (base)	0% (assumed = INF list)	14,472,408	87,502,697	0	9,257,578	0
60 days	0.63% (interpolated)	14,563,584	87,415,730	-86,967	10,463,612	2,427,264
90 days	1.27% (one test-Y1 order)	14,656,207	87,327,383	-175,314	10,463,612	3,664,052
120 days (Y1)	3.00% (measured, REP p4)	14,906,580	87,088,569	-414,128	10,463,612	4,968,860

[~] Only the 120-day premium is measured on our own volumes (1.6995 / 1.0815 against list 1.65 / 1.05). The 90-day figure is one 795,600-unit test-Y1 order; 60 days is a straight line between; cash = list is an assumption (\$28 says price depends on volume and terms and gives no numbers). The trough is positive at every setting, so terms are a pure price question: cash saves 414,128 against Year 1 if list holds and at least 239,000 (the 90-day floor) if it does not, and it measures the cash price we will need for Years 3-4. The 1.2M of January cash the longer terms preserve is not needed (section 4.2).

Finance lever	Setting	Year-2 cost / income	Net income	Delta vs base	January trough	Market
Loan: keep 19,500,000	prepay 0	interest 1,950,000	85,747,697	-1,755,000	28,952,578	[!]
Loan: prepay half	prepay 9,750,000	interest 975,000 + fee 97,500	86,625,197	-877,500	19,105,078	[~]
Loan: prepay all (base)	prepay 19,500,000	fee 195,000	87,502,697	0	9,257,578	[+]
Credit line 10,000,000	keep	200,000 (1% + 1%, REP p5)	87,302,697	-200,000	unchanged	[!]
Credit line 0 (base)	drop	0	87,502,697	0		[+]
Training department ON	722,000	turnover 306,000	86,840,697	-662,000		[!]
Training department OFF (base)	0	turnover 366,000 (4 x	87,502,697	0		[+]

		24,000 + 9 x 30,000)				
Fixed-term deposit	0 / 5M / 10M / 15M	0 / 125,000 / 250,000 / 375,000	-	-	19.3M / 14.3M / 9.3M / 4.3M	[~]

[+] Prepaying 19.5M at 1% against a 10% coupon is 1,755,000 a year and the January trough stays at 9.26M with the deposit on top. [~] The base assumes the prepayment executes on 2 January with the instalment and stops Year-2 interest (Y1 charged a full year of interest on the draw year, REP p5, so the simulator accounts by year). If interest is instead charged on the opening balance, Year 2 loses 1,950,000 and the prepayment still saves 1,950,000 of Year 3-4 interest for 195,000: open question 1.

[!] Training: Y1 paid 722,000 and turnover cost was 306,000, the same as test Y1 with the department off; \$31-\$32 give it no effect other than turnover, so the net saving is 662,000. The 10M deposit is the single line to cut if the team wants a bigger January cushion; even a forced early withdrawal costs 1% (INF p2), less than the 250,000 it earns.

Form 1 — Commercial distribution

Field	Y1 A	Y1 B	Y1 E	Y2 A	Y2 B	Y2 E	Source
Sales managers	6	6	0	6	6	0	\$6; Doc 06: managers reset every year, re-enter. E stays out (paragraph below)

Why not E in Year 2. E's potential is 110,100,000 units (\$3 x \$2) and the two participants that entered sold 98,165 between them: Co2 took 92,329 units on 12 platforms and a full media line, Co1 lost 10,887,540 on 48 platforms for 5,836 units (survey 3, 23, 27). Local brands J, K and L sit on E's three largest segments (\$17) and spend \$8,000,000 each on media (test survey 12). Our own test-Y2 entry cost \$14.24 per unit on 112,814 units (Doc 06 Form 1: \$1,606,400 for managers, media, POP and vehicles). With S supply-capped a million units below domestic demand and 17.6M of H to sell at home, there is no unit to send. Surveys 17 and 18 (60,000) are on Form 4 so the Year-3 decision is made on E prices and volumes rather than on this paragraph.

Form 2 — Logistics

Field	Y1 A	Y1 B	Y1 E	Y2 A	Y2 B	Y2 E	Source
Platforms	12	12	60	12	12	12 (form min)	\$10 allowed values; Y1 warehouse 0.0295/unit at 12; E: \$13 forbids zero, 12 is the floor
Transport to platforms:	3	2	10	2	2	1 (form	Section 2.7

number of vehicles						min)	
Transport to platforms: load capacity (kg)	24,000	24,000	12,000	24,000	24,000	1,000 (form min)	\$10 max; 0.0044/unit at full load (-- vehicles)
Transport to retailers: number of vehicles	30	20	30	24	18	1 (form min)	Section 2.7
Transport to retailers: load capacity (kg)	10,000	10,000	1,500	10,000	10,000	500 (form min)	\$12 max

[!] The Y1 sheet left the worksheet template's E defaults (60 platforms, 10 x 12,000 and 30 x 1,500 vehicles) in place with no E price. It cost nothing (REP: no E line anywhere; Doc 02 brief §2.6) but survey 27 shows rivals "60 platforms in E" for us. The cells to change are exactly the five E cells above; if the interface accepts 0 in them, enter 0, otherwise the minimums. E is exited by the blank E price (§15), as in Y1.

2.7 Fleet sizing for Year-2 volume

Volume by market at the base case (\$ 6,675,000 with the unserved Y1 demand assigned to A; H at Y1 market shares): **A 11,309,770 • B 10,412,624**, 21.7M in total against 20.4M shipped in Y1 with no stockout flag on 3 + 2 trunk and 30 + 20 last-mile vehicles.

Leg	Y1 fleet	Y1 charge	Y2 fleet	Y2 charge	Load per vehicle, base / high case (H 21M)	Evidence
Trunk A	3 @ 24,000	79,200	2 @ 24,000	52,800	24,000,000 capacity: 2.12x / 1.52x	One vehicle moves 12,000,000 a year (\$10 arithmetic, Doc 06 §2.3, four flags)
Trunk B	2 @ 24,000	52,800	2 @ 24,000	52,800	2.30x / 1.65x	Test Y2 stocked out at 1.05x; Y1 cleared at 3.5x
Last mile A	30 @ 10,000	540,000	24 @ 10,000	432,000	471,240 / 657,659 units per vehicle	Proven: 343,866 (Y1 A), 504,016 (Y1 B); proven

						failure: 1,217,870 (test Y2 B)
Last mile B	20 @ 10,000	360,000	18 @ 10,000	324,000	578,479 / 807,320	same
Total	55 vehicles	1,032,000	46 vehicles	861,600	saving 170,400	Charges at the measured 50% of the \$10/\$12 formula (03 fix 4)

[~] The model's geometric estimate (`plan_platform_to_retailer`) asks for 29 A / 16 B at 93 and 61 hours per route; Doc 06 and the model itself say not to plan on it. The cut stops at loads the fleet has been seen to carry (B carried 504,016 per vehicle in Y1) with the high case still under the only measured failure point. A stockout on this leg costs an emergency premium, not a sale (\$8: test Y2 paid 1,527,519 over the tariff), so the next 8 vehicles (144,000) are the cheapest insurance on the sheet if the team prefers Y1's counts.

Form 3 — Human resources

Field	Y1	Y2	Source
Level I	126,000	126,000	\$29 range 108,000-150,000
Level II	84,000	84,000	\$29 range 72,124-99,170
Level III	60,000	60,000	\$29 range 51,100-72,124
Level IV	35,000	35,000	\$29 range 30,050-42,075
Operator	13,500	13,500	\$30 range 12,000-16,830
Specialist	24,000	24,000	\$30 range 21,000-29,450
Recruitment and training department	Yes	No	\$32: 722,000; Y1 turnover 306,000 with it on = test Y1 with it off. Net saving 662,000 (section 2.4)

Office payroll 2,818,000 (`office_payroll()` , REP p5). Production headcount at 10 S + 2 H lines x 3 shifts: 306 specialists + 1,098 operators = 1,404, plus 24 SMED specialists (2 teams x 3 shifts x 4, \$27); Y1 ran 3,144 (`production_headcount()`).

Sales reps per zone	Y1 S	Y1 T	Y2 S	Y2 T	Source
A	4	4	4	4	Unchanged; S is supply-capped and H in A is a positioning/promotion question (surveys 13, 14, 15)
B	5	3	5	3	Unchanged; B holds 55.8% of H

Sales force salary	Y1 fixed / variable	Y2 fixed / variable	Source
A	27,000 / 0.60	27,000 / 0.60	\$13 range 21,000-42,000 / 0.6-1.2
B	27,000 / 0.60	27,000 / 0.60	same

[~] \$13-\$15 do not link reps to shelf space or channel coverage: reps are hired per zone for channels S and T (\$13, synoptic 2.3), shelf space is a separate objective the retailer negotiates (\$21), and managers handle channel G (\$6). One extra rep per B zone in one channel is $6 \times 27,000 = 162,000$ fixed. It is a cheap test, but not for S (no supply to sell) and not needed for H in B (55.8% share); survey 14 (24,000) shows rivals' rep counts first. Sales network fixed cost stays 3,192,000; commission estimate 740,022.

Form 4 — Market research surveys

Survey	Content	Cost	Y1	Y2	Why now
1	Sales of S by market	3,000	Yes	Yes	bundle
2	Sales of H by market	3,000	Yes	Yes	bundle
3	Sales of S and H by company and market	36,000	Yes	Yes	shares
4	Sales by company, market and channel	72,000	No	Yes	Where Co2 and Co4 sell H in A (supermarkets?)
7	Positioning of S brands	59,900	Yes	Yes	campaign IDs are meaningless without it
8	Positioning of H brands	59,900	Yes	Yes	Co2 sits on our (9,1)
10	Coverage of advertising by company and market	59,900	No	Yes	rival media weight
11	Number of advertising insertions by company	36,000	No	Yes	rival media weight
13	POP investment by company,	59,900	No	Yes	the lever we tripled blind

	market, channel				
14	Sales managers and reps by company and market	24,000	No	Yes	salesforce weight
15	Shelf space achieved by company and channel	72,000	No	Yes	did 22% land, and what did the winners get
17	Average overseas prices	24,000	No	Yes	Year-3 E gate (Doc 06 build path)
18	Total overseas sales by A/B/E companies	36,000	No	Yes	Year-3 E gate
19	Demand for normal units by company and market	110,000	No	Yes	rivals' unserved demand: is S capacity-capped for everyone, and where was our S demand
20	Rivals' retail prices	0	Yes	Yes	free
21	Rivals' campaigns	0	Yes	Yes	free
22	Rivals' promotion types	0	Yes	Yes	free
23	Net income by company	1,000	Yes	Yes	bundle
24	Industry average production cost	10,000	No	Yes	is 5.15 / 2.23 competitive
26	Industry balance sheets	10,000	No	Yes	rivals' cash and stock: can they add capacity
27	Platforms by company and market	3,000	Yes	Yes	bundle
28	Shifts by company	1,000	Yes	Yes	implied lines, with survey 3

5, 6, 9, 12, 16, 25	-	-	No	No	12 (E domestic advertising, 36,000) was read in the test round: \$8M each
Total		680,600	166,800	680,600	research_cost(...); 453,800 of additions + 60,000 E gate

[+] 513,800 more than Y1 against 40,990,251 of H stock the Year-1 surveys could not explain and a million units of S demand we could not place. The whole menu is 1,003,400; the six left out are E promotional detail and salary averages.

Form 5 — Product S: price, margin, shelf, promotions

Market	Channel	Y1 MSRP	Y2 MSRP	Margin (€)	Shelf (%)	Promotion	Number	Det
A	G	14.50	15.00	1.90	18	No promotion	0	0
A	S	14.50	15.00	1.90	18	No promotion	0	0
A	T	14.50	15.00	1.90	18	No promotion	0	0
B	G	14.50	15.00	1.90	18	No promotion	0	0
B	S	14.50	15.00	1.90	18	No promotion	0	0
B	T	14.50	15.00	1.90	18	No promotion	0	0
E	G	0.00	0.00	0.00	0	No promotion	0	0

[!] No S promotion anywhere: output 6,675,000 is 1,002,440 below last year's demand and \$48 rejects a promotion the supply cannot serve. The brief's B promotion test moves to the year S supply exceeds A demand.

POP advertising (€)	A G	A S	A T	B G	B S	B T	E G	Source
S (Y1 and Y2)	10,000	10,000	10,000	10,000	10,000	10,000	0	DEP p5; \$19 cap 500,000

								per channel
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Advertising campaign S	A	B	E	Source
Campaign (Y1 and Y2)	18	18	0	DEP p5; 52.7% A share, positioning (1,8)

Advertising repetitions S (Y1 = Y2)	A: No.	A: Cost (€)	B: No.	B: Cost (€)	E: No.	Source
DR Display	0	0	0	0	0	
PR Press	4	60,000	4	60,000	0	\$15 rate 15,000; 312.5 per reach point (--media)
RA Radio	0	0	0	0	0	
SM Social media	0	0	0	0	0	500 per reach point, dearest (--media)
TV Television	8	144,000	8	144,000	0	\$15 rate 18,000; 300 per reach point, cheapest
Total	12	204,000	12	204,000	0	advertising("A", {"TV": 8, "PR": 4}): 79.2% net reach

Form 6 — Product H: price, margin, shelf, promotions

Market	Channel	Y1 MSRP	Y2 MSRP	Margin (€)	Shelf (%)	Y2 Promotion	Number	Det
A	G	6.50	7.50	1.00	22	No promotion	0	0
A	S	6.50	7.50	1.00	22	Price reduction	3	0.5
A	T	6.50	7.50	1.00	22	No promotion	0	0

B	G	7.50	7.50	1.00	22	No promotion	0	0
B	S	7.50	7.50	1.00	22	Price reduction	3	0.5
B	T	7.50	7.50	1.00	22	No promotion	0	0
E	G	0.00	0.00	0.00	0	No promotion	0	0

[~] "Detail" is read as the euro reduction per unit (test Y2's direct bonus entered Detail 2 and cost 0.60 = 1.20 / 2 per promo unit, Doc 05 Part 2), so 0.50 is the price cut. If the interface labels Detail as a budget, enter 0.50 x expected promo units: A 137,405 (274,811 units), B 524,296 (1,048,592 units). Open question 3.

POP advertising (€)	A G	A S	A T	B G	B S	B T	E G	Source
H (Y1 and Y2)	15,000	15,000	15,000	15,000	15,000	15,000	0	DEP p6

Advertising campaign H	A	B	E	Source
Campaign (Y1 and Y2)	19	19	0	DEP p6; survey 8: Co2 runs 19 on the same (9,1)

Advertising repetitions H (Y1 = Y2)	A: No.	A: Cost (€)	B: No.	B: Cost (€)	E: No.	Source
DR Display	0	0	0	0	0	
PR Press	3	45,000	3	45,000	0	\$15
RA Radio	0	0	0	0	0	
SM Social media	0	0	0	0	0	
TV Television	5	90,000	5	90,000	0	\$15
Total	8	135,000	8	135,000	0	advertising("A", {"TV": 5, "PR": 3}): 79.2% net reach

Media total S + H: **678,000**. POP total: **150,000**. Both unchanged: the surveys on Form 4 say whether Y3 media should move, and moving it now would confound them.

Form 7 — Production

Field	Y1 H	Y1 S	Y2 H	Y2 S	Source
Lines previously installed	0	0	22	10	REP p6; should be pre-filled, verify
Lines installed this year	22	10	0	0	\$22: 32-line cap reached
Total lines	22	10	22	10	
Active lines	22	10	2	10	Section 2.3; \$22 lines can be switched off, never removed
Stock in units of RM	0	0	0	298,350	REP p4; should be pre-filled, verify
Stock in units of FP	0	0	17,628,699	0	REP p3-p4; should be pre-filled, verify
Units purchased of RM	32,818,500	6,265,350	3,446,019	6,578,235	H = 2 x 3 x 557,500 x 1.02 x 1.01; S = 10 x 3 x 222,500 x 1.02 x 1.01 - 298,350

[!] Raw material arithmetic. Used units were exactly 1.0200 x produced in both Y1 and the test (REP p4: 5,967,000 / 5,850,000; 32,818,500 / 32,175,000), and output per line-shift has landed on the measured figure both years, so the 1% buffer covers only one thing: 222,500 / 557,500 were measured with 2 modules serving 32 lines, and 12 lines on 2 modules may run slightly above it (the model's uncalibrated 93.70% would give 224,880 / 562,200, i.e. +71,400 S and +28,200 H units). Under-ordering costs +15% on the shortfall (\$28); the buffer ties up 68,085 S + 34,119 H units (~151,000) that carry into Year 3. Raw material cost at list on cash terms: S 10,854,088 + H 3,618,320 = **14,472,408**.

Production options	Y1	Y2	Source
Number of shifts	3	3	\$23; plant-wide; depreciation 12% (\$35)
Preventive maintenance	Yes	Yes	\$25, 500,000; part of the measured 195,000 / 222,500 per line-shift
Pokayoke previous years	No	No	
Pokayoke	No	No	\$26: 720,000 x 32 installed lines = 23,040,000 for +4.55%

SMED	Y1	Y2	Source
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SMEDs previous years	0	0	
To acquire	0	2	\$27: 3,000,000 each, year 2 onwards; test Y2 DEP p6 ran 2 / 2 / 2 and delivered 222,500 / 557,500
Total SMED	0	2	
Active	0	2	
Teams assigned to SMED per shift	0	2	\$27: teams of 4 specialists at the specialist salary; 576,000 a year at 24,000

Unit cost the forecast expects at these settings: **S 5.1491 • H 2.2305** (section 1 decomposition); the H stock sells at the \$33 weighted average **2.3101**. Test Y2 on preventive + SMED at 120-day terms came in at S 5.1762 / H 2.0605 (H on 0.9379 material bought earlier), so 5.15 at cash terms is inside the measured range.

Form 8 — Finance

Field	Y1	Y2	Source
Loan requested	26,000,000	0	Section 4: OCN 9,621,161 against 52,305,895 available
Loan prepayment	0	19,500,000	\$42, INF p2: 1% penalty (195,000) against 1,950,000 of interest; trough 9,257,578 after it (4.2)
Financial investment	0	0	\$44: same 2.5% as the deposit line below; one instrument keeps the form simple
Fixed-term deposits	0	10,000,000	\$44, INF p2: 2.5% = 250,000; sized to keep the January trough above 9M (section 2.4 deposit row)
Line of credit requested	10,000,000	0	\$43: cost 200,000 undrawn in Y1 (REP p5); the January trough no longer depends on Year-2 sales
Customer payment terms	60 Days	60 Days	\$38 shortest option; receivable 30,875,094 at year end (2/12 of net revenue)
Factoring	No	No	Nothing to fund
Max. amount to be discounted	--	--	
Payment term for suppliers S	120 Days	Cash	Section 2.4: 120 days cost 3% (REP p4 1.6995 vs 1.65); cash saves up to 414,128 and measures the cash price
Payment term for suppliers H	120 Days	Cash	same; 3,618,320 of purchases

Investment in securities 1-3	0 / 0 / 0	0 / 0 / 0	\$45
Sale of shares 1-3	No / No / No	No / No / No	

4. Cash check

4.1 The \$41 test the simulator runs

Component	€	Source
80% x new fixed assets 6,000,000 (SMED)	4,800,000	\$41 item 1
50% x (media 678,000 + POP 150,000 + managers 600,000 + rep fixed 2,592,000 + overhead 600,000 + research 680,600)	2,650,300	\$41 item 2
15% x raw material 14,472,408	2,170,861	\$41 item 3
OCN	9,621,161	operating_cash_needed()
Available cash 2 January: 46,902,208 - 6,500,000 due - 15,380,390 payables + 27,284,077 receivables	52,305,895	available_cash_jan2() (\$41); matches the brief's figure
Same after the 19,500,000 prepayment + 195,000 penalty and the 10,000,000 deposit	22,610,895	Whether the simulator nets these first is unknown; it passes either way
Headroom	+12,989,734	financing_gap() -> loan_needed 0

[+] No forced loan under either reading. The Y1 test was the same shape (OCN 52.5M vs 54.0M) and the simulator borrowed nothing beyond the request (03 \$a).

4.2 The real trough (monthly, before financing)

Doc 06 \$4.2 method, even spreading. 2 January: -6,500,000 instalment, -19,695,000 prepayment and penalty, -6,000,000 SMED, -10,000,000 deposit. Year-1 receivables 27,284,077 arrive half in January and half in February (60 days); Year-1 payables 15,380,390 leave a quarter a month January-April (120 days). Year-2 net revenue 185,250,561 arrives from month 3 at 1/12 a month; raw material 14,472,408 is paid from month 1 (cash terms); conversion cost (labour, power, repairs, SMED teams), general expenses less depreciation, promotions and logistics leave at 1/12 a month.

Month-end	Jan	Feb	Mar	Apr	May	Dec
Cash	9,257,578	13,807,948	20,153,826	26,499,704	36,690,680	108,027,50

Lever	Effect on the January trough	Cost
Without the 10,000,000 deposit	19,257,578	-250,000 of income

Prepay half instead of all	19,105,078	-877,500
120-day supplier terms	10,463,612	-414,128
Keep the 10,000,000 credit line	unchanged (undrawn)	-200,000
No SMED	15,551,370	-7,492,530

[+] The January trough is the year's low point and is independent of Year-2 sales: nothing sold in Year 2 is collected before March. The 20% overdraft (\$42) needs a 9.3M error in a month whose inflows and outflows are all Year-1 numbers already on the balance sheet. [~] The one unmodelled item is whether the simulator books the prepayment on 2 January (open question 1); if it books it on 31 December the trough is 28.8M and the interest saving moves to Years 3-4.

4.3 Year-2 result and the year-end position

Base case: revenue 212,980,455; net revenue 185,250,561; gross profit 113,956,440; operating income 87,447,697; **net income 87,502,697**. Year-end: cash 108,292,278, deposit 10,000,000, receivables 30,875,094, inventory 13,838,518 (H 5,926,305 units at 2.3101 + 68,085 S and 34,119 H raw-material units), payables 0, loan 0. Available cash for the Year-3 OCN test: **149,167,372** (cash + deposit + receivables, nothing due).

Cash reconciliation (ties to the euro): 46,902,208 + 87,502,697 net income + 7,200,000 depreciation - 3,591,017 receivables growth + 27,658,779 inventory release - 15,380,390 payables paid - 6,000,000 SMED - 6,500,000 instalment - 19,500,000 prepayment - 10,000,000 deposit = 108,292,277.

[~] For scale: Year 1 earned 67,823,192 with 40,990,251 of production cost parked in stock and 3,060,000 of financial expense. This plan releases 27.7M of that stock into cash, carries 55,000 of net financial income, and still leaves 5.9M H units and a million S units of demand for Year 3.

5. A6 pre-submission checklist, ticked against this sheet

Check	Status	Where
Price, retailer margin and shelf space in every cell we sell in	Yes	Forms 5-6: 12 A/B cells; E blank
Advertising campaign selected	Yes	18 S / 19 H in A and B
Lines activated, not just installed	Yes	Form 7: 10 S + 2 H active; 20 H deliberately off
Production forecast <= capacity at the measured rate	Yes	S 6,675,000 / H 3,345,000 at 222,500 / 557,500 per line-shift
Raw material covers production plus scrap	Yes	x1.02 x1.01 both products; S opening stock netted
No logistics field blank or zero	Yes	Form 2; E at form minimums (or 0 if accepted)
Every zone covered by a manager or wholesaler	Yes	6/6 managers in A/B; E not operated

Promotion volumes within supply	Yes	H: 20,973,699 available vs ~1.3M promo units; S: no promotion (\$48)
OCN <= available cash, or loan deliberately requested	Yes	9,621,161 <= 22,610,895 after prepayment and deposit
Surplus cash on deposit	Yes	10,000,000 fixed-term; January trough 9,257,578
Surveys 20, 21, 22 + the 11,000 bundle ordered	Yes	Form 4, total 680,600
SMED requested only from year 2; Poka Yoke not requested	Yes	Form 7: 2 modules, 2 active, 2 teams per shift; PY No
Sales managers entered (they reset every year)	Yes	Form 1: 6 / 6 / 0
Loan prepayment entered, no new loan, no credit line	Yes	Form 8
E template defaults cleared	Yes	Form 2 E cells; Forms 5-6 E rows 0.00 / 0

6. Open questions to confirm on the interface or with the Director (max 5)

#	Question	If the answer is bad
1	Is a loan prepayment executed on 2 January with the instalment (\$42, \$47) so that no Year-2 interest accrues on the 19,500,000, or is interest charged on the opening balance regardless?	Prepay anyway: Year 2 loses 1,950,000 of interest it would have paid regardless, the penalty is 195,000, and Years 3-4 save 1,300,000 + 650,000
2	How many production lines can one SMED duplicate module serve? Test Y2 measured 2 modules on 32 active lines at 222,500 / 557,500; Year 2 runs 12 active lines. Is one module enough, and must teams per shift equal modules?	Keep 2 modules (base). If the Director confirms one module covers 12 lines, buy 1 (saves 3,000,000 of cash and 360,000 of depreciation), keep 2 teams
3	Promotion form semantics for "Price reduction": is Detail the euro cut per unit (0.50) or a budget? Test Y2's direct bonus entered Detail 2 and cost 0.60 per promo unit	If a budget: enter 0.50 x expected promo units per cell (A supermarkets 137,405; B supermarkets 524,296)
4	Why did the Year-1 REP print 501,093 of unserved H demand alongside 17,628,699 of H stock and no logistic stockout flag (REP p3-p4)? Channel or shelf-space cap?	If shelf-capped, raise H shelf space from 22% toward the survey-15 reading of the sell-out rivals in Year 3; do not change it blind now
5	If cash goes negative mid-year, does the simulator liquidate a fixed-term deposit (1% fee,	Cut the deposit to 5,000,000 (trough 14.3M, income 125,000) rather than

INF p2) or charge the 20% overdraft (\$42)?

keep the credit line at 200,000

7. Where the brief did not survive the rulebook

Brief item	What the rulebook or the Year-1 numbers say
"Add 4 S lines (10 -> 14), capex per --lines"	§22: 32-line cap, already reached by 10 + 22. Replaced by 2 SMED modules (6,000,000): S 6,675,000, not 8,190,000. [!]
"Keep 14.50 in A, test a B promotion at 14.50"	S output is 1,002,440 below Y1 demand; §48 rejects promotions beyond supply. Base moves S to 15.00 (+3,176,230 unless demand falls 18.8%); B promotion deferred. [~]
"Apply the market growth constant if the rulebook gives one"	None exists: §3 gives per-capita consumption at t0 and no growth rate; INF changed nothing (03 §c). No growth is assumed anywhere in this sheet
"Re-size vehicles for ~25M units"	Base volume is 21.7M (S capped at 6.7M); fleet cut to 46 vehicles, not below loads the fleet has carried
"Surveys ... also 12, 17, 18 if we want the E decision for Y3"	17 and 18 bought (60,000); 12 skipped, its content was read in the test round
"Deposit surplus at 2.5% if the model/rulebook allows"	§44 and INF p2: fixed-term deposits at 2.5%, 1% early-withdrawal fee. 10,000,000 sized to the January trough
"Supplier terms 60/90/120 vs the trough"	Trough clears at every setting; cash terms chosen on price (only the 120-day premium is measured on our volumes; cash = list is the assumption). [~]

Sources

- Scenario MMT39 EN v231.pdf — §3, §6, §8, §10-§15, §17-§23, §25-§30, §31-§38, §41-§45, §47-§49 (summarised in docs/01-company-and-market-dossier.md and sources-md/synoptic-overview.md)
- day 2/year 1 results/DIR0114SI02083TE3Y1REP.md — units by channel p2-p3, production and stockout flags p4, P&L p4-p5, balance sheet p6
- day 2/year 1 results/DIR0114SI02083TE3Y1DEP.md — Year-1 form values (the Y1 columns above), form order
- day 2/year 1 results/DIR0114SI02083TE3Y1INFen_US.md — rates, 4-year maturity, 1% prepayment penalty, 2.5% deposits
- day 2/year 1 results/DIR0114SI02083TE3Y1INV.md — surveys 1, 2, 3, 7, 8, 20-23, 27, 28
- day 2/year 1 results/analysis/01-results-vs-plan.md — opening position, cost structure, 3% terms premium, training verdict
- day 2/year 1 results/analysis/02-competitor-brief.md — rival prices, promotions, implied lines, penetration, E
- day 2/year 1 results/analysis/03-model-vs-actual.md — the twelve corrections applied by hand
- day 2/year 2 results/DIR0114SI02082TE3Y2DEP.md and ...REP.md — test-Y2 SMED entries (2 / 2 / 2) and output per line-shift 222,500 / 557,500

- docs/05-test-year2-postmortem-and-graded-run-plan.md — Part 4 SMED measurement, conversion cost 3.4276 / 1.1046, third-module verdict, promotion Detail evidence
- docs/06-graded-year1-plan-2.1.md — structure, \$4.2 trough method, E entry cost, build-path gates
- model/mmt39.py — capacity(), standard_cost(), production_headcount(), office_payroll(), vehicle_annual_cost(), plan_platform_to_retailer(), operating_cash_needed(), available_cash_jan2(), financing_gap(), research_cost(), promotion_cost_per_unit(), advertising(), receivables(), SCENARIO
- plan_y2.py (scratchpad) — every base-case, sensitivity and alternative figure above